第十一讲 (2026.10.10)

Thm H —— Hilbert 空间

H 可分 $\Leftrightarrow$ H 有至多可数个 O.N.B.

Pf "$\Leftarrow$"

设 $\{e_k\}_{k=1}^{\infty}$ 为 O.N.B.

令 $M \overset{def}{=} span^{\mathbb{Q}}(\{e_k\}_{k=1}^{\infty})$

$$\overset{def}{=} \left\{ \sum_{k=1}^{n} \lambda_k e_k : \lambda_k \in \mathbb{Q}+i\mathbb{Q},\ \begin{matrix} k=1,\cdots,n \\ n\in\mathbb{N} \end{matrix} \right\}$$

Claim 1 M 可数

$$\varphi : M \longrightarrow \bigcup_{I \subset \{e_k\}_{k=1}^{\infty},\ \#I<\infty} (\mathbb{Q}+i\mathbb{Q})^{\#I}$$

$I = \{e_{n_k}\}_{k=1}^{n}$

$$\sum_{k=1}^{n} \lambda_k e_{n_k} \longmapsto (\lambda_1, \cdots, \lambda_n)$$

为一一对应

Claim 2 $M \overset{dense}{\subset} H$

$\forall\, x \in H$,

$$x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$$

$\forall\, \varepsilon > 0$, $\forall\, n$, $\exists\, \alpha_n \in \mathbb{Q} + i\mathbb{Q}$. s.t.

$$\left| \langle x, e_n \rangle - \alpha_n \right| < \frac{\varepsilon}{2^{n+1}}$$

$\Rightarrow \forall\, N$,

$$\left\| \sum_{n=1}^{N} \langle x, e_n \rangle e_n - \sum_{n=1}^{N} \alpha_n e_n \right\|^2$$

$$= \sum_{n=1}^{N} \left| \langle x, e_n \rangle - \alpha_n \right|^2$$

$$< \frac{\varepsilon^2}{4}$$

而当 N 充分大时

$$\left\| x - \sum_{n=1}^{N} \langle x, e_n \rangle e_n \right\| < \frac{\varepsilon}{2}$$

$$\Rightarrow \left\| x - \underbrace{\sum_{n=1}^{N} \alpha_n e_n}_{\in M} \right\| < \varepsilon$$

HW: 举一个不可分 Hilbert 空间的例子.

Def $(X_1, \langle\cdot,\cdot\rangle_1)$, $(X_2, \langle\cdot,\cdot\rangle_2)$

如果 $\exists\, T: X_1 \to X_2$ 线性同构 s.t.

$$\langle Tx, Ty\rangle_2 = \langle x, y\rangle_1, \quad \forall x, y \in X_1$$

则称 X_1 与 X_2 作为内积空间同构.

记为 $X_1 \simeq X_2$.

Thm (i) n 维 Hilbert 空间 $H \simeq \mathbb{K}^n$

(ii) 无穷维可分 Hilbert 空间 $H \simeq l^2$

Pf: (ii) 设 $\{e_k\}_{k=1}^{\infty}$ 为 H 的 O.N.B.

定义 $T: H \to l^2$

$$x \longmapsto \{\langle x, e_k\rangle\}_{k=1}^{\infty}$$

1° 线性

2° 保距

$$\|Tx\|_{l^2} = \left(\sum_{k=1}^{\infty} |\langle x, e_k\rangle|^2\right)^{1/2}$$

$$= \|x\| \qquad (\text{by P.I.})$$

3° 单射 (by 2°)

4° 满射

$\forall\, a \in \ell^2$,

$$\left\| \sum_{k=n}^{m} a_k e_k \right\|^2 = \sum_{k=n}^{m} |a_k|^2 \to 0$$

as $m, n \to \infty$

$\Rightarrow \exists\, x \in H$ s.t.

$$\left\| \sum_{k=1}^{n} a_k e_k - x \right\| \to 0 \quad \text{as} \quad n \to \infty$$

$$x = \sum_{k=1}^{\infty} a_k e_k$$

II

$$\langle x, e_k \rangle = a_k, \quad k = 1, 2, \cdots$$

$\Rightarrow Tx = a$

5° $\langle x, y \rangle = \left\langle \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k, \sum_{j=1}^{\infty} \langle y, e_j \rangle e_j \right\rangle$

$$= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \langle x, e_k \rangle \overline{\langle y, e_j \rangle} \underbrace{\langle e_k, e_j \rangle}_{\delta_{kj}}$$

$$= \sum_{k=1}^{\infty} \langle x, e_k \rangle \overline{\langle y, e_k \rangle}$$

$$= \langle Tx, Ty \rangle_{\ell^2}$$

$$\mathbb{T} = \{ e^{2\pi i t} : t \in \mathbb{R} \}$$ (單位圓周) ⑤

- 設 $F : \mathbb{T} \to \mathbb{C}$

$$f(t) \stackrel{def}{=} F(e^{2\pi i t}) \quad , \quad t \in \mathbb{R}$$

$\Rightarrow$ f 為 $\mathbb{R}$ 上周期函數

- $F \leftrightarrow f$, $\mathbb{T} \leftrightarrow [-\frac{1}{2}, \frac{1}{2})$

- 考慮

$$e_k(t) \stackrel{def}{=} e^{2\pi i k t} \quad , \quad t \in [-\frac{1}{2}, \frac{1}{2})$$

$$k = 0, \pm 1, \pm 2, \cdots$$

(三角函數系.)

- Thm $\{e_k\}_{k=1}^{\infty}$ 為 $L^2(\mathbb{T})$ 的 O.N.B.

設 $f \in L^2(\mathbb{T})$

$$\hat{f}(k) \stackrel{def}{=} \int_{-\frac{1}{2}}^{\frac{1}{2}} f(t) e^{-2\pi i k t} \, dt$$

$$= \langle f, e_k \rangle \quad , \quad k \in \mathbb{Z}$$

- $$f(x) \sim \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{2\pi i k x} = \sum_{k=-\infty}^{\infty} \langle f, e_k \rangle e_k$$

Thm' $\forall f \in L^2(\mathbb{T})$

$$\| S_N(f) - f \|_2 \to 0 \quad \text{as} \quad N \to \infty$$

其中

$$(S_N f)(x) \overset{\text{def}}{=} \sum_{k=-N}^{N} \hat{f}(k)\, e^{2\pi i k x}$$

Thm' $\Longleftrightarrow$ $\{e_k\}_{k\in\mathbb{Z}}$ 是 $L^2(\mathbb{T})$ 的 O.N.B.

$\Longleftrightarrow$ $\left(\{e_k\}_{k\in\mathbb{Z}}\right)^{\perp} = \{0\}$

$\Longleftrightarrow$ $\overline{\operatorname{span}\{e_k\}_{k\in\mathbb{Z}}} = L^2(\mathbb{T})$

$(M^{\perp})^{\perp} = \overline{\operatorname{span}(M)}$ (Ex. 1.6.5)

$\{\text{三角多项式}\} \overset{\text{dense}}{\subset} L^2(\mathbb{T})$

$$(S_N f)(x) = (f * D_N)(x)$$

其中

$$D_N(t) \overset{\text{def}}{=} \sum_{k=-N}^{N} e^{2\pi i k t} = \frac{\sin[(2N+1)\pi t]}{\sin(\pi t)}$$

称为 Dini 核.

$$\sigma_N f \overset{\text{def}}{=} \frac{1}{N+1} \sum_{k=0}^{N} S_k f \qquad (\text{Cesàro 求和法})$$

$$= f * F_N$$

其中

$$F_N(t) \overset{\text{def}}{=} \frac{1}{N+1} \sum_{k=0}^{N} D_k(t)$$

$$= \frac{1}{N+1} \frac{\sin^2[(N+1)\pi t]}{\sin^2(\pi t)}$$

称为 Fejér 核.

1° $\int_{-\frac{1}{2}}^{\frac{1}{2}} F_N(t)\,dt = 1$

2° $\forall \delta > 0$

$$\lim_{N\to\infty} \int_{\delta<|t|<\frac{1}{2}} F_N(t)\,dt = 0$$

[记 $\delta < |t| < \frac{1}{2}$ 时,

$$0 \le F_N(t) \le \frac{1}{N+1} \frac{1}{\sin^2(\pi\delta)} \quad]$$

Lem (Minkowski 积分不等式)

设 $1 \le p < \infty$,

$$\left\| \int_Y f(\cdot, y)\,dy \right\|_p \le \int_Y \|f(\cdot, y)\|_p\,dy$$

Pf

$$\left\{ \int_X \left| \int_Y f(x,y)\, dy \right|^p dx \right\}^{1/p}$$

$$\leq \int_Y \left\{ \int_X |f(x,y)|^p dx \right\}^{1/p} dy$$

Thm $\forall f \in L^2(\Pi)$

$$\| \sigma_N f - f \|_2 \to 0 \quad \text{as} \quad N \to \infty$$

Pf $\| \sigma_N f - f \|_2$

$$= \left\{ \int_{-\frac{1}{2}}^{\frac{1}{2}} \left| \int_{-\frac{1}{2}}^{\frac{1}{2}} [f(x-t) - f(x)]\, F_N(t)\, dt \right|^2 dx \right\}^{1/2}$$

$$= \left\| \int_{-\frac{1}{2}}^{\frac{1}{2}} [f(\cdot - t) - f(\cdot)]\, F_N(t)\, dt \right\|_2$$

Minkowski

$$\leq \int_{-\frac{1}{2}}^{\frac{1}{2}} \| f(\cdot - t) - f(\cdot) \|_2 \, F_N(t)\, dt$$

$$= \int_{|t| \leq \delta} + \int_{\delta < |t| < \frac{1}{2}}$$

$$= I_1 + I_2$$

$I_1 < \varepsilon/2$ 当 δ 充分小.

(由积分的绝对连续性)

$$I_2 \leq 2\|f\|_2 \int_{\delta<|t|<1/2} F_N(t)\,dt$$

$< \varepsilon/2$ 当 N 充分大.